ECOM90009 Quantitative Methods for Business

Workshop 2 (Week 3)

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Introduction

Zheng Fan

- Know more about myself: zhengfan.site
- Ph.D student in Economics at Unimelb

If you need any help, don't be shy

- Weekly consultation & Ed discussion board (your first priority)
- Email: `jnlye@unimelb.edu.au` for subject related issues
- Consult Stop 1 for special consideration
- Email: `fan.z@unimelb.edu.au`

Before asking any questions, make sure you have gone through the materials on Canvas and existing posts on Ed discussion board!

Introduction

- Measures of variability
- Percentiles
- Graphical visualisation

Files downloaded from Canvas

- `W2-2-1.xlsx`

Question 1

Indicate whether each statement below is true or false. Justify your answer.

(a) When a distribution is symmetric, the mean, median and the mode are all equal.

(b) The difference between the largest and smallest values in an ordered array is called the range.

(c) The value of the standard deviation may be either positive or negative, while the value of the variance will always be positive.

(d) A courier company is reviewing their delivery times. The following descriptive statistics relates to this courier company. The mean time for package delivery is 5 hours, the first quartile is 4 hours, the third quartile is 17 hours. This means that the 25% of the deliveries take more than 17 hours.

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Note:

Symmetric distribution does not imply normal distribution.

True

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True

Question 1(c)

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Population variance and standard deviation

$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2 \qquad \sigma = \sqrt{\sigma^2} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2}$$

Sample variance and standard deviation

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \mu)^2 \qquad s = \sqrt{s^2} = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \mu)^2}$$

False

Question 1(d)

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Percentiles

The pth percentile of a set of observations is the value for which at most $p\%$ of observations lie below that value, and at most $(100 - p)\%$ of observations lie above it.

True

Question 2 (same data as Workshop 1 Question 4)

According to the OECD (Organisation for Economic Cooperation and Development), student skills relevant for the labour market contribute to a level of a country's well-being, and higher skill levels are indicators of higher well-being. The data in the excel spreadsheet `W1.xlsx` contains data on the 2020 student skill scores for 37 OECD countries. Summarize this data using an appropriate graphical display and report summary statistics including the mean, median and mode. Determine the locations and values of the first and third quartiles. Summarize your results.

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- How do we use Excel to obtain the summary statistics (mean, median and mode)?
- For the first and third quartiles, how do we determine their locations?

Histogram in Excel

Excel data analysis Histogram defaults at (left, right]

For example, if we set bins as 410, 420, 430, 440, . . . , 520, 530

The class intervals will be

- $x \leq 410$
- $410 < x \leq 420$
- $420 < x \leq 430$
- $430 < x \leq 440$
- $\vdots$
- $520 < x \leq 530$
- $x > 530$

Determining the Number of Classes

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- The appropriate number of classes to use is ultimately a subjective decision
- Table 4.3 in the textbook suggests that where number of observations is less than 50, the number of classes is $5 - 7$
- Alternatively, Sturge's formula:

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Approximate location of any percentile p

$$L_p = (n + 1)\frac{p}{100}$$

Summarising result

- Mean = 489.3
- Median = 496
- Mode = 504
- First quartile = 478
- Third quartile = 507

Distribution of Student Skills:

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Distribution of Student Skills:

```
is unimodal and skewed to the left.  The mean (489.3) is slightly
lower than the median (496) because the distribution of data
is skewed to the left.  Three lowest student skills' scores are
from Colombia, Mexico and Turkey.  The middle 50% of country's
student skill scores are between 478 and 507.
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Question 3

On a very cold morning on the 28th January 1986, the US space shuttle "Challenger" exploded soon after take off. All seven crew members who were on board were killed. A group of engineers had unsuccessfully attempted to halt the launch. They argued that it was unsafe to proceed given the likelihood of malfunctions occurring when the temperature was below freezing. Why were the engineers unable to convince their senior managers to recommend to NASA that the launch not proceed? In this question we will investigate this by constructing some simple scatter plots using the data provided in the sheet named Q3 in W2.xlsx.

Question 4

You are one of three partners in a firm.

Excluding you and your partners, the firm has 100 employees, and the total earnings of these 100 employees last year was \$7 million. There is little variability in these employees' earnings. You and your two partners are paid \$750,000 each. Your firm has made profits of \$900,000 which are to be paid in dividends to the partners.

One of your partners has suggested summarising this information in the following way for the company's annual report:

- Average earnings in the firm \$89,806
- Average profit of partners \$300,000

Do you think this is a reasonable suggestion?

Discussion

100 employees + you + 2 other partners = 103 people

- Total earnings of the 100 employees = \$7,000,000 (little variability)
- You + 2 other partners each earn \$750,000
- Profits of \$900,000 paid in dividends to the partners

(suggested) Summarising of information:

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Do you think this is a reasonable suggestion?

$\therefore$ Huge difference in the mean annual earnings and it's unreasonable to suggest the information according to your partner's proposal.

`On average, Bill Gates and I are both billionaires.`

Thanks for your attention!

Let me know if there is any questions!